

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing(01CT1513)	Aim: Program to generate attached signals and to make comprehensive documents along with program, output, conclusion.	
Open Ended Problem: 1	Date: 19/07/2026	Enrollment No: 92400133138

Aim:

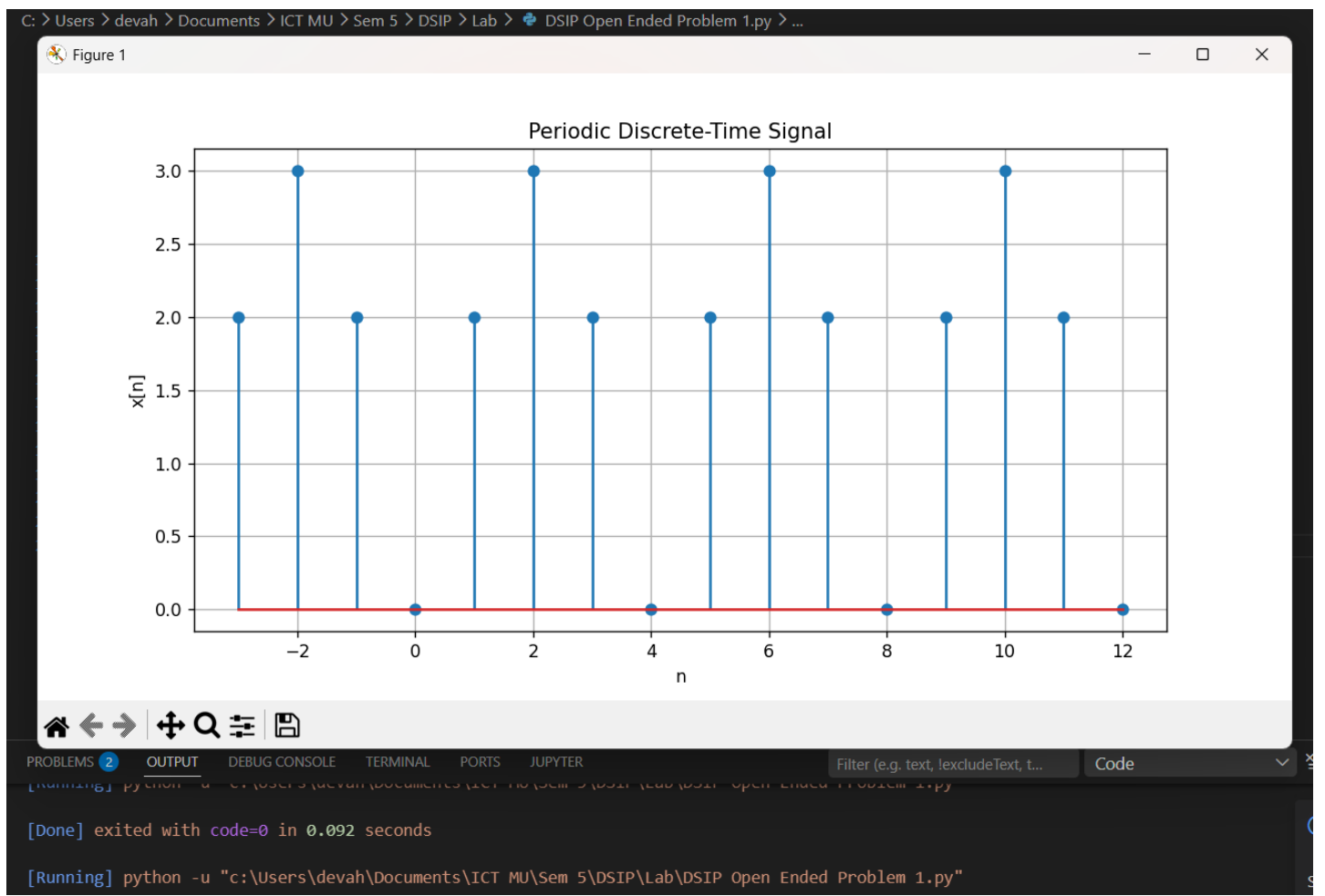
To generate and plot the given periodic discrete-time signal using Python and understand the concept of periodic discrete signals.

Code:

```
C: > Users > devah > Documents > ICT MU > Sem 5 > DSIP > Lab > DSIP Open Ended Problem 1.py > ...
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Sample values
5  n = np.arange(-3, 13)
6
7  # One period of the signal
8  pattern = [2, 3, 2, 0]
9
10 # Generate periodic signal
11 x = np.array([pattern[(i + 3) % 4] for i in n])
12
13 # Plot
14 plt.figure(figsize=(10,5))
15 plt.stem(n, x)
16
17 plt.title("Periodic Discrete-Time Signal")
18 plt.xlabel("n")
19 plt.ylabel("x[n]")
20 plt.grid(True)
21
22 plt.show()
```

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Output:



Conclusion:

The given periodic discrete-time signal was successfully generated and plotted using Python. The signal repeats its pattern [2, 3, 2, 0] after every 4 samples, confirming that it is a periodic discrete-time signal with a fundamental period of 4. The experiment demonstrates how periodic sequences can be efficiently generated using arrays and the modulo operator in Python.

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Aim:

To generate and plot the given **staircase (piecewise constant) discrete-time signal** using Python and understand the concept of piecewise signals in Digital Signal Processing.

Code:

```

C: > Users > devah > Documents > ICT MU > Sem 5 > DSIP > Lab > DSIP Open Ended Problem 1.py > ...
29 import numpy as np
30 import matplotlib.pyplot as plt
31
32 # Sample indices
33 n = np.arange(-2, 7)
34
35 # Initialize signal
36 x = np.zeros_like(n)
37
38 # Generate staircase signal
39 x[(n >= 0) & (n < 1)] = 1
40 x[(n >= 1) & (n < 2)] = 2
41 x[(n >= 2) & (n < 4)] = 3
42 x[(n >= 4) & (n < 5)] = 2
43 x[(n >= 5) & (n < 6)] = 1
44
45 # Plot
46 plt.figure(figsize=(8,5))
47 plt.stem(n, x)
48
49 plt.title("Generated Staircase Signal")
50 plt.xlabel("n")
51 plt.ylabel("x[n]")
52 plt.xticks(np.arange(-2,7))
53 plt.yticks(np.arange(0,4))
54 plt.grid(True)
55
56 plt.show()

```

PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
JUPYTER

Filter (e.g. text, !excludeText, t...

```

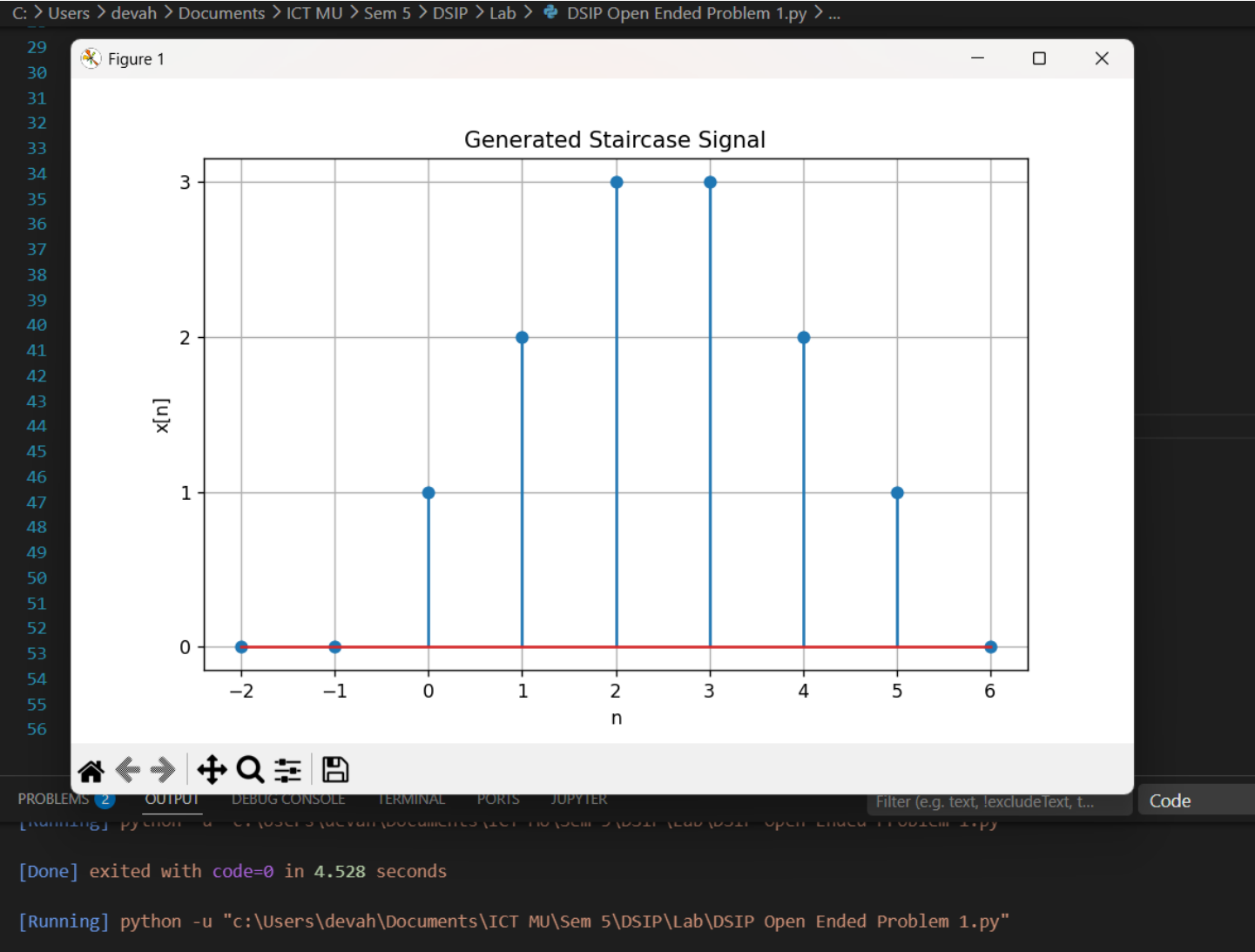
[Done] exited with code=0 in 4.528 seconds

[Running] python -u "c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\DSIP Open Ended Problem 1.py"

```

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Output:



Conclusion:

The given staircase signal was successfully generated and plotted using Python. The signal starts with zero amplitude, increases in discrete steps from 1 to 3, remains at the maximum value for two samples, and then decreases symmetrically back to zero. The experiment demonstrates how piecewise constant discrete-time signals can be implemented using logical indexing in NumPy and visualized using Matplotlib's stem() function, which accurately represents discrete signal samples.

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Aim:

To generate and plot the given piecewise discrete-time signal using Python and understand the implementation of signals composed of constant and ramp sections.

Code:

```

C: > Users > devah > Documents > ICT MU > Sem 5 > DSIP > Lab > DSIP Open Ended Problem 1.py > ...

62 import numpy as np
63 import matplotlib.pyplot as plt
64
65 # Sample values
66 n = np.arange(-4, 5)
67
68 # Initialize signal
69 x = np.zeros_like(n, dtype=float)
70
71 # Constant section
72 x[(n >= -2) & (n < 0)] = -2
73
74 # Ramp section
75 x[(n >= 0) & (n <= 1)] = 2 * n[(n >= 0) & (n <= 1)]
76
77 # Constant positive section
78 x[(n > 1) & (n <= 2)] = 2
79
80 # Plot
81 plt.figure(figsize=(8,5))
82 plt.stem(n, x)
83
84 plt.title("Piecewise Signal")
85 plt.xlabel("n")
86 plt.ylabel("x[n]")
87 plt.grid(True)
88
89 plt.show()

```

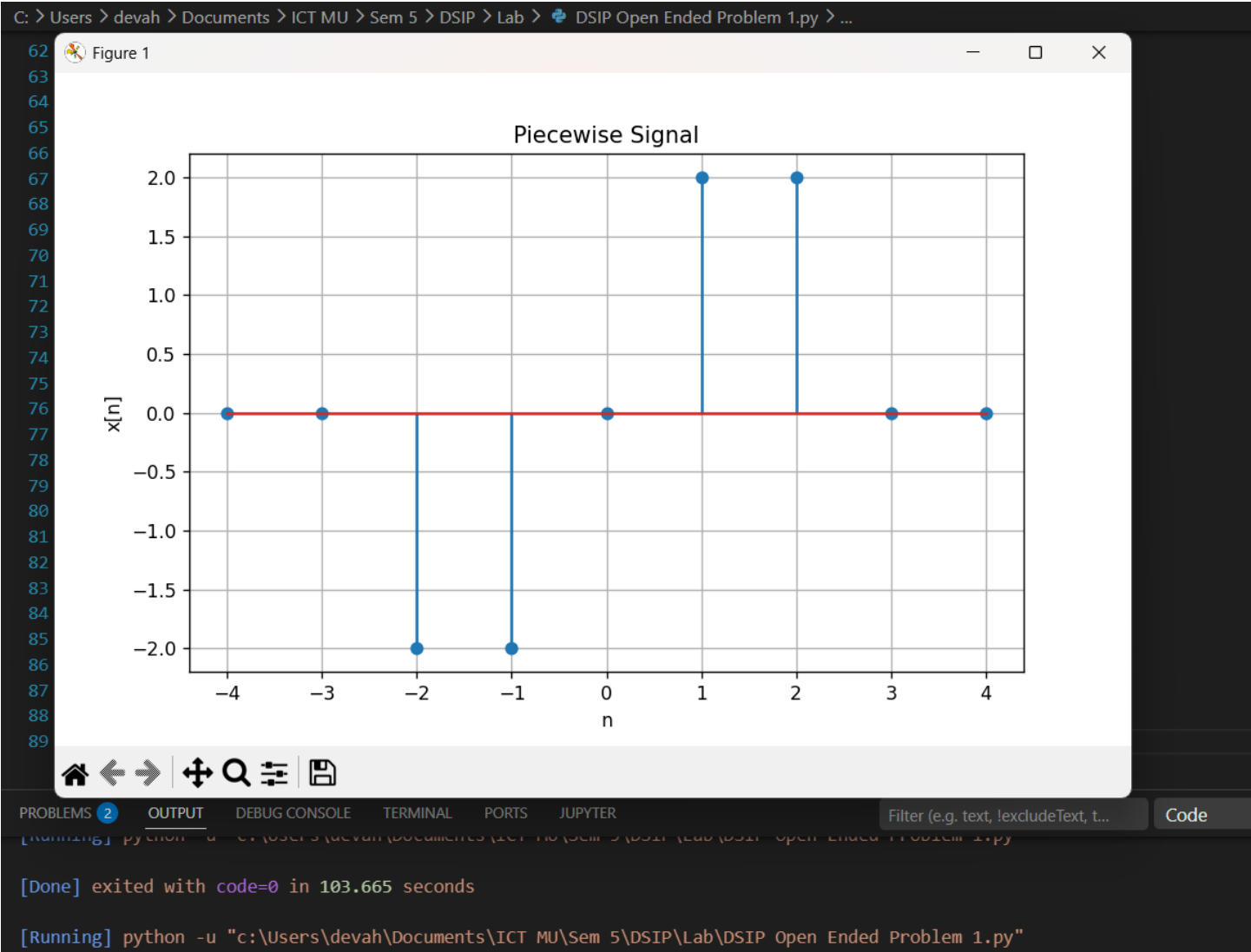
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER Filter (e.g. text, !excludeText, t...

[Done] exited with code=0 in 103.665 seconds

[Running] python -u "c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\DSIP Open Ended Problem 1.py"

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Output:



Conclusion:

The given piecewise discrete-time signal was successfully generated and plotted using Python. The signal consists of a negative constant section, a linearly increasing ramp, and a positive constant section, accurately matching the given waveform. This experiment demonstrates how different elementary signal segments can be combined to create complex discrete-time signals using logical indexing in NumPy and visualized using Matplotlib.

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Aim:

To generate and plot the discrete-time signal represented using unit step functions in Python.

Code:

```

C: > Users > devah > Documents > ICT MU > Sem 5 > DSIP > Lab > DSIP Open Ended Problem 1.py > ...
94
95 import numpy as np
96 import matplotlib.pyplot as plt
97
98 # Sample indices
99 n = np.arange(-2, 12)
100
101 # Unit step signals
102 u = (n >= 0).astype(int)
103 u3 = (n >= 3).astype(int)
104 u7 = (n >= 7).astype(int)
105
106 # Given signal
107 x = u - u3 - 5*u7
108
109 # Plot
110 plt.figure(figsize=(9,5))
111 plt.stem(n, x)
112
113 plt.title("x[n] = u[n] - u[n-3] - 5u[n-7]")
114 plt.xlabel("n")
115 plt.ylabel("x[n]")
116 plt.grid(True)
117
118 plt.show()

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

Filter (e.g. text, !excludeText, t...) Code

[Done] exited with code=0 in 167.248 seconds

[Running] python -u "c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\DSIP Open Ended Problem 1.py"

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Output:



Conclusion:

The output verifies the behavior of shifted and scaled unit step functions, producing a piecewise discrete-time signal with amplitudes 0, 1, 0, and -5 over different intervals. The experiment demonstrates how complex signals can be synthesized using elementary unit step functions, a fundamental concept in Digital Signal Processing.

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Aim:

To generate and plot the discrete-time signal represented using unit impulse functions in Python.

Code:

```

C: > Users > devah > Documents > ICT MU > Sem 5 > DSIP > Lab > DSIP Open Ended Problem 1.py > ...
122
123 import numpy as np
124 import matplotlib.pyplot as plt
125
126 # Sample indices
127 n = np.arange(-5, 6)
128
129 # Initialize signal
130 x = np.zeros_like(n)
131
132 # Generate impulses
133 x[n == 0] = 1          #  $\delta[n]$ 
134 x[n == 1] += 3         #  $3\delta[n-1]$ 
135 x[n == -1] += 5        #  $5\delta[n+1]$ 
136
137 # Plot
138 plt.figure(figsize=(8,5))
139 plt.stem(n, x)
140
141 plt.title("x[n] =  $\delta[n] + 3\delta[n-1] + 5\delta[n+1]$ ")
142 plt.xlabel("n")
143 plt.ylabel("x[n]")
144 plt.grid(True)
145
146 plt.show()

```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER Filter (e.g. text, lexcludeText, t...

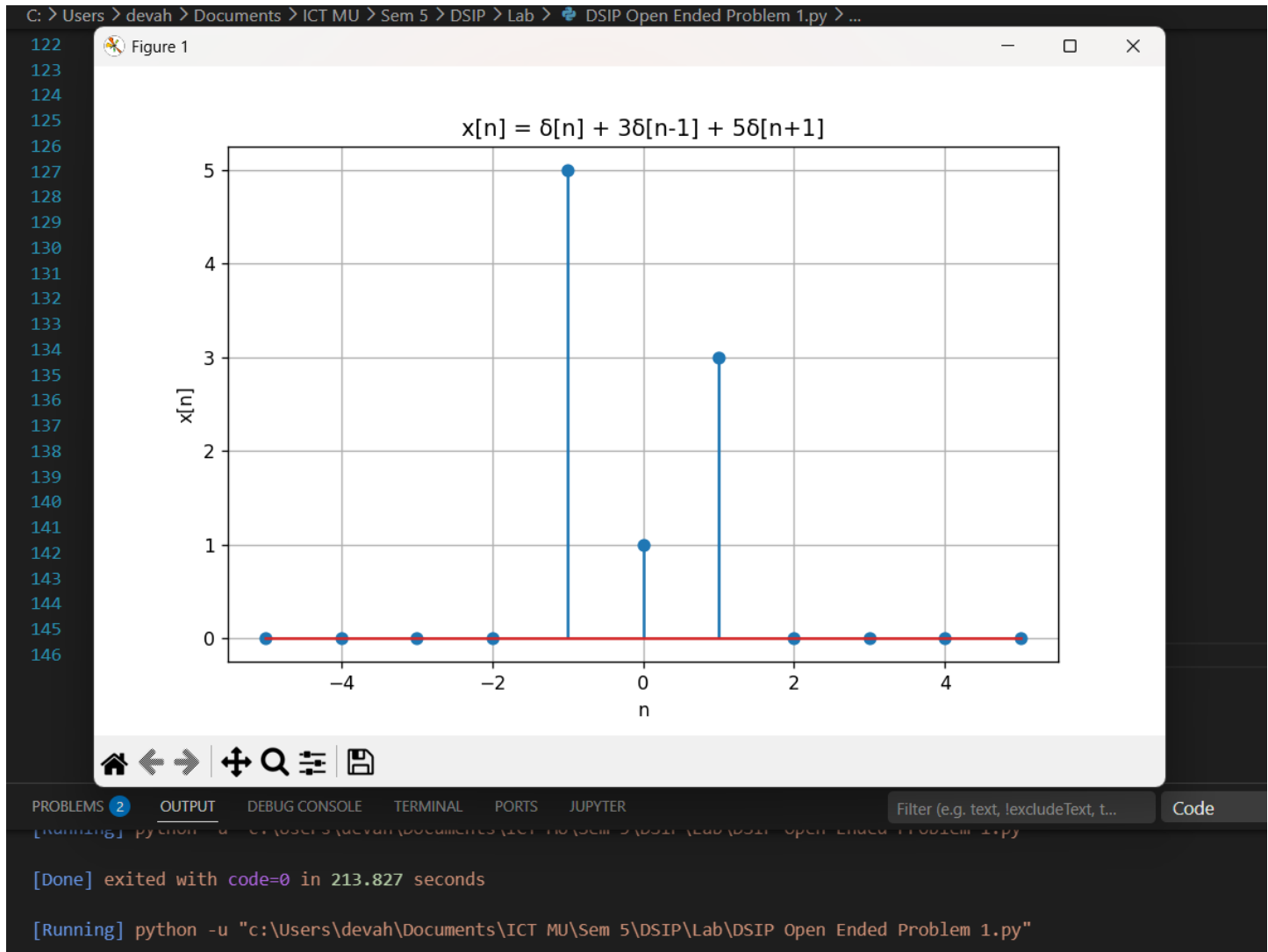
[Running] python -u "c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\DSIP Open Ended Problem 1.py"

[Done] exited with code=0 in 213.827 seconds

[Running] python -u "c:\Users\devah\Documents\ICT MU\Sem 5\DSIP\Lab\DSIP Open Ended Problem 1.py"

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Output:



Conclusion:

The output contains three impulses located at $n = -1$, $n = 0$, and $n = 1$ with amplitudes 5, 1, and 3, respectively. This experiment demonstrates how shifted and scaled unit impulse functions can be combined to construct discrete-time signals, reinforcing a fundamental concept in Digital Signal Processing.